

2. Newton's second law of motion tells us that this rate of change of velocity (centrifugal acceleration) must be equal to the gravitational force, F_{grav} exerted by the Sun on the planet, divided by the planet's mass, M_{planet}; In other words, $2\pi C/P^2 = F_{grav}/M_{planet}$.
3. Newton's gravitational law tells us that the gravitational force F_{grav} is proportional to the Sun's mass M_{Sun} times the planet's mass M_{planet} divided by the square of the planet's orbital circumference. Stated as an equality rather than a proportionality, $F_{grav} = 4\pi^2 G M_{Sun} M_{planet}/C^2$. Here G is Newton's constant of gravitation, equal to 6.670×10^{-20} kilometer^3 per second^2 per kilogram, or equivalently 1.327×10^{11} kilometers^3 per second^2 per solar mass.
4. By inserting this expression for the gravitational force F_{grav} into Newton's second law of motion (Step 2 above), we obtain $2\pi C/P^2 = 4\pi^2 G M_{Sun}/C^2$. By then multiplying both sides of this equation by $C^2/2\pi$, we obtain $C^3/P^2 = 2\pi G M_{Sun}$.

Thus, Newton's laws of motion and gravity explain—in fact they enforce—the relationship discovered by Kepler: C^3/P^2 is the same for all planets; it depends only on Newton's gravitation constant and the Sun's mass.

As an illustration of the power of the laws of physics, the above manipulations not only explain Kepler's discovery, they also offer us a method to weigh the Sun. By dividing the final equation in Step 4 by $2\pi G$, we obtain an equation for the Sun's mass, $M_{Sun} = C^3/(2\pi G P^2)$. By inserting into this equation the circumference C and period P of any planet's orbit as measured by astronomers and the value of Newton's gravitation constant G as measured in Earth-bound laboratories by physicists, we infer that the mass of the Sun is 1.989×10^{30} kilograms.

62 ["Weber lectured . . . his every class."] Document 39 in ECP-1; Document 2 in Einstein and Marić (1992).

63 [And since the aether . . . at rest in absolute space,] In this chapter, I ignore the speculations by some physicists in the late nineteenth century that in the vicinity of the Earth the aether might be dragged along by the motion of the Earth through absolute space. There in fact was strong experimental evidence against such dragging: If, near the Earth's surface, the aether was at rest with respect to the Earth, then there should be no aberration of starlight; but aberration due to the Earth's motion around the Sun was a well-established fact. For a brief discussion of the history of ideas about the aether, see Chapter 6 of Pais (1982); for more detailed discussions, see references cited therein.

64 [Albert Michelson . . . had invented.] The technology of Michelson's time was not